CHAPTER FOUR

MEMORY

&

FORGETTING



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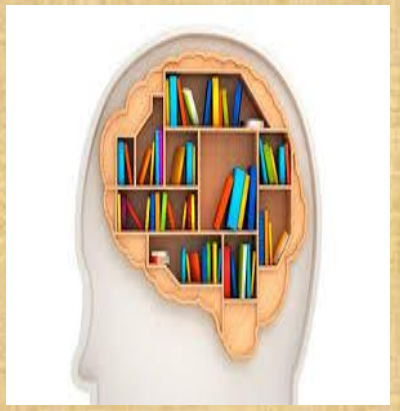
Meaning of Memory

- **Memory:** is the ability to retain information



over time

- It's the capacity to encode, store and retrieve information

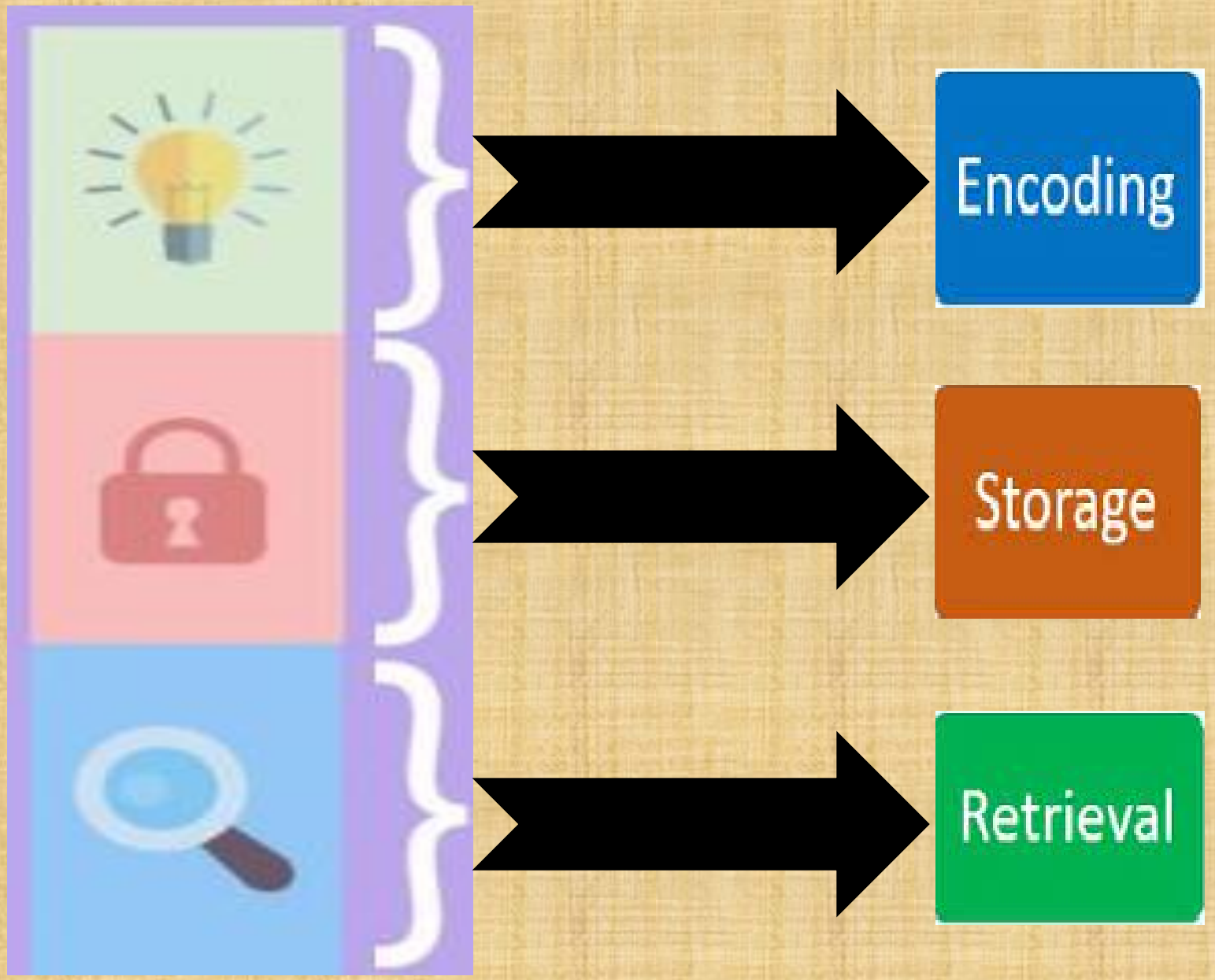


- It is the ability to remember past events or previously learned information or skill



- It is a complex mental system.
- That involves multiple stages of information processing

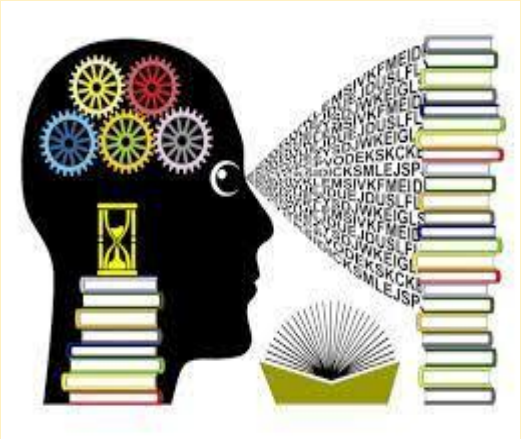
Memory Processes



Memory Processes

- ***Encoding***: transforming information into a form that can be entered and retained in the memory system.
  - Integrating information in to the memory system.
  - It could be through association or other means

E.g Associating George ⟹ George Washington



Cont …

- ***Storage***: retaining information in memory so that it can be used at a later time.

  - Some physiological change in the brain must take place ⟹ **Consolidation**

  - Preserving information for latter use



- ***Retrieval***: recovering information

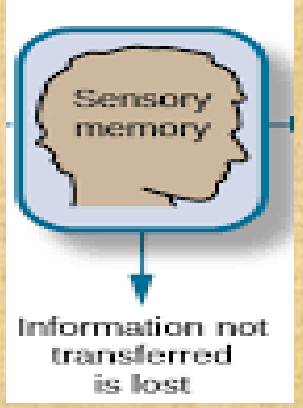
stored in memory so that we are consciously aware of it.

Eg. Calling George by his name when you meet



Types of Memory

Sensory Memory/ Register



- SM is the first step in the memory process.
- It has very rich and detailed information

  - But information stays for only a very short time

- Function—holds information long enough to be processed for basic physical characteristics
- Capacity— very large
  - It can hold many items at once
- Duration—very brief retention of images

Cont…

Iconic memory briefly holds visual information during eye blink.

- SM divided into **two** types:
  - **Iconic memory**–holds visual information for about a quarter of a second or more



- As soon as you shift your attention, the information disappears

- **Echoic memory**– is an auditory information,
  - Holds auditory information from 1-2 seconds



Echoic memory briefly holds sounds.

Short Term or Working Memory

- **STM**: Sometimes referred to as ***working memory***.

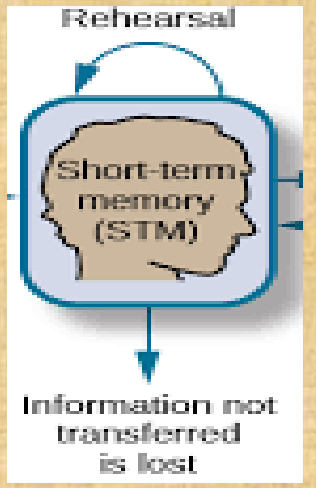
  - holds a limited amount of information

  - On average seven items – for a limited period of time i.e. up to 30 seconds

- Function—conscious processing of information
  - where information is actively worked on.

- Capacity—limited storage (holds 7±2 items)

- Duration—brief storage (about 30 seconds)



Characteristics of STM

- ***It is active***: informat^{n} stays as long as the person is consciously examining and manipulating it.
- ***Rapidly accessed***: informat^{n} easily disappears unless it is accessed in LTM
- ***Preserves the temporal sequence*** : Informat^{n} is memorized as it is orderly arranged
- ***Limited in capacity***: People can hold on average 7$\pm$ 2 pieces of informat^{n} at a time.

Controlling Informat^{n} in STM

- **Rehearsal**: repeating informat^{n} again and again – maintenance rehearsal

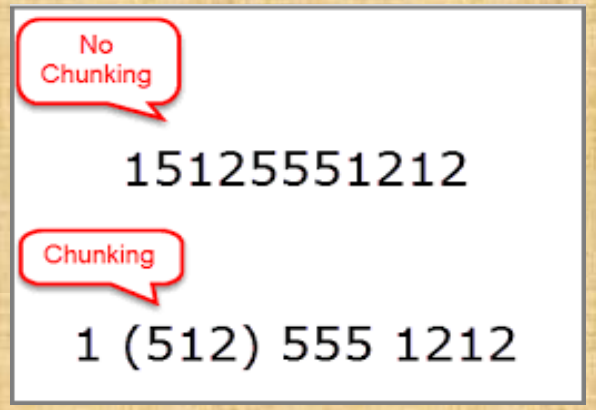
Maintenance rehearsal is not a very effective encoding process.

- Elaborative rehearsal is important in transferring informat^{n} to LTM (involves creating meaning, associating & interpreting concepts.

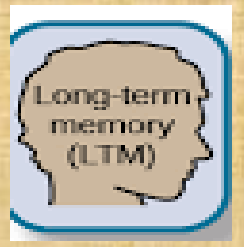
- **Chunking**: a process of breaking down large Informat^{n} in to its smaller parts

Which is easier to remember?

- **4 8 3 7 9 2 5 1 6**
- **483 792 516**



Long term memory



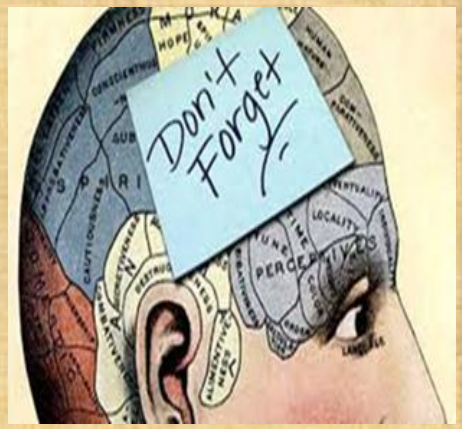
- A relatively permanent type of memory
  - Holds huge amount of information for a long time
- Has unlimited capacity to store information
- Has a potential to retrieve or remember inf^{n} in the future

➢ **Types of long term memory**

➥ **Explicit/declarative memory**: refers to knowledge that can be consciously brought to mind and "declared."

- It is conscious memory of facts and events

- It can be **semantic** or **episodic**



Cont…

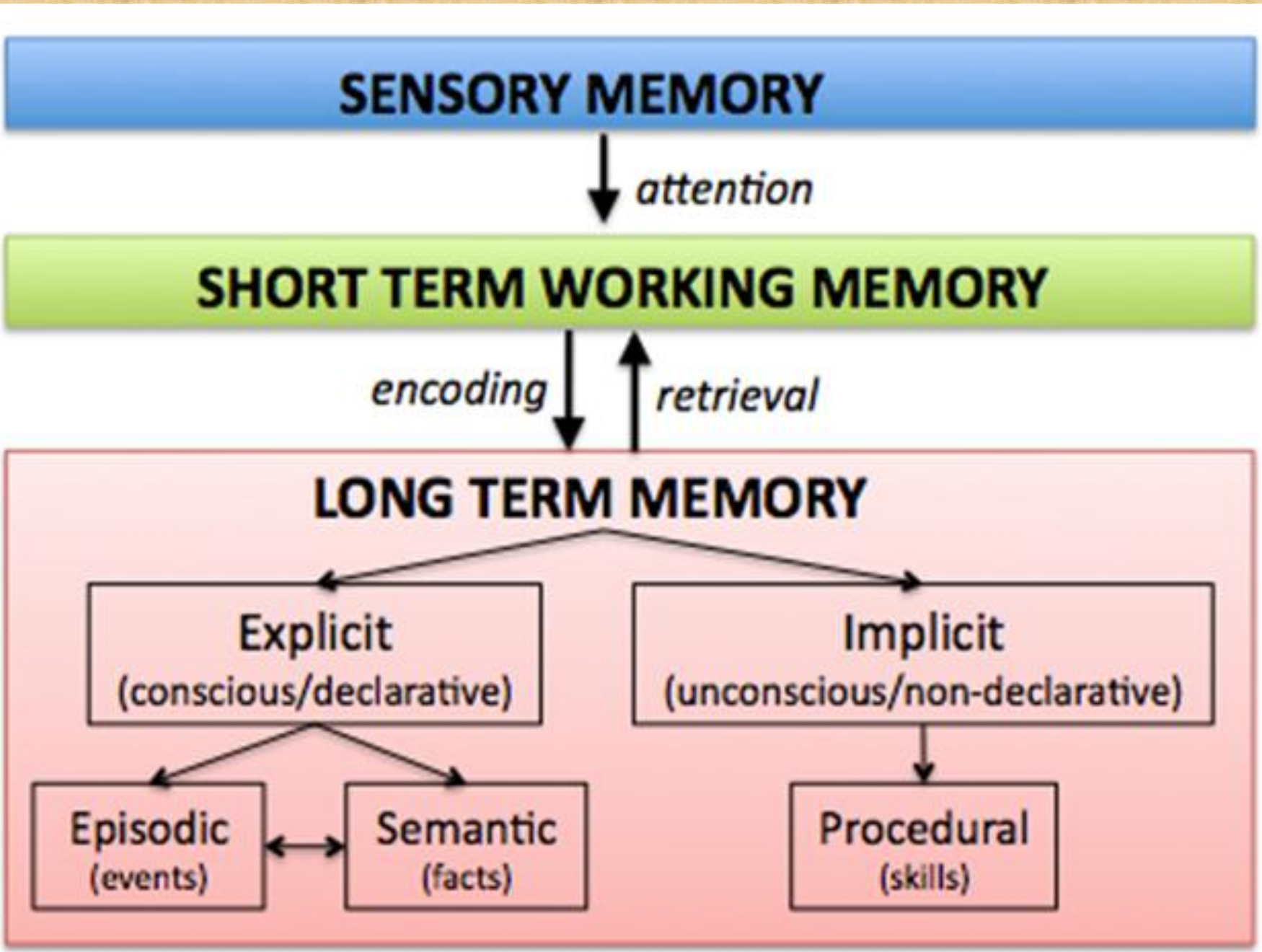
- ***Semantic Memory***: refers to general world knowledge or facts, such as:

  - Three systems of memory: H_2O, NaCl, H_2SO_4

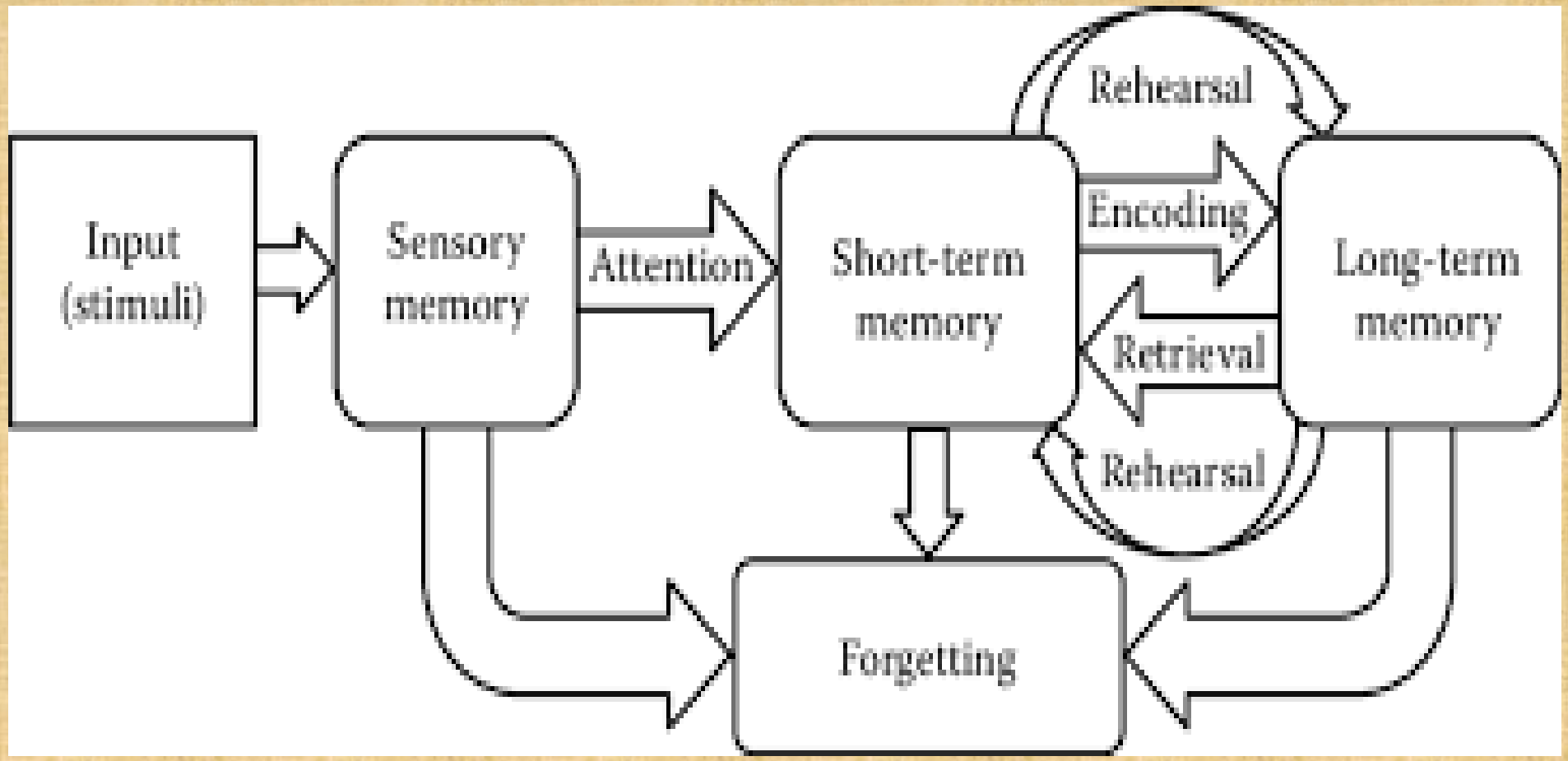
- ***Episodic Memory***: consists of memories of **particular** episodes or events from personal experience.

➪ **Implicit memory**: is based on experience with **out consciously** recollected

- It ***cannot be verbalized*** or ***consciously recollected***
- It can be **procedural memories** that requires applying skills of:
  - Playing tennis,
  - Riding a bicycle & typing


SENSORY MEMORY
attention
SHORT TERM WORKING MEMORY
encoding
retrieval
LONG TERM MEMORY
Explicit
(conscious/declarative)
Implicit
(unconscious/non-declarative)
Episodic
(events)
Semantic
(facts)
Procedural
(skills)

Cont…

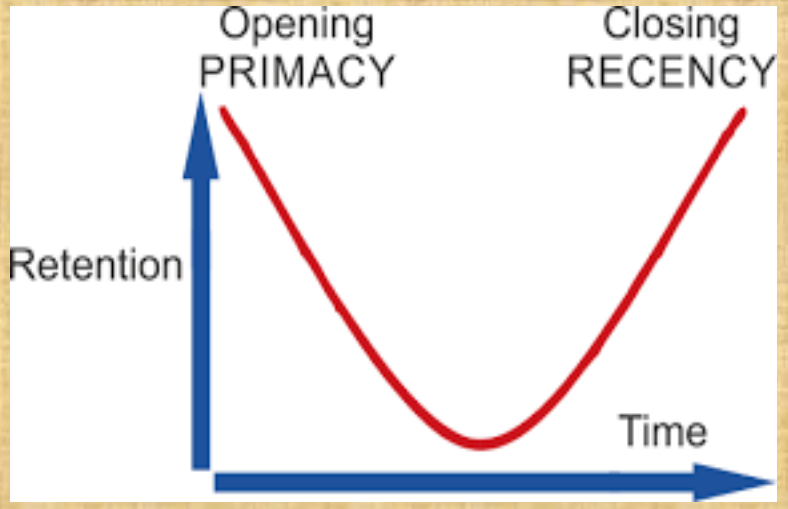

Input
(stimuli)
Sensory
memory
Attention
Short-term
memory
Rehearsal
Encoding
Retrieval
Rehearsal
Long-term
memory
Forgetting

The serial position effect

- Ebbinghaus studied memory by using senseless syllables. Such as:

(VEM, BOZ, FUW etc.)

- He found that items in **the middle are difficult to remember** than those at **the beginning** and **end** – called **serial position effect**



- Better memory of items at the **beginning** of a list is called **the primacy effect**. And

- Better memory for items at **the end** of a list is called the **recency effect**.

Factors Affecting Memory

- **Ability to retain**: depends upon good memory traces left in the brain by past experiences
- **Good health**: A person with good health can retain the learnt material better than a person with poor health
- **Age of the learner**: Youngsters can remember better than the aged.
- **Maturity**: Very young children cannot retain and remember complex material
- **Will to remember**: Willingness to remember
- **Intelligence**: More intelligent person will have better memo

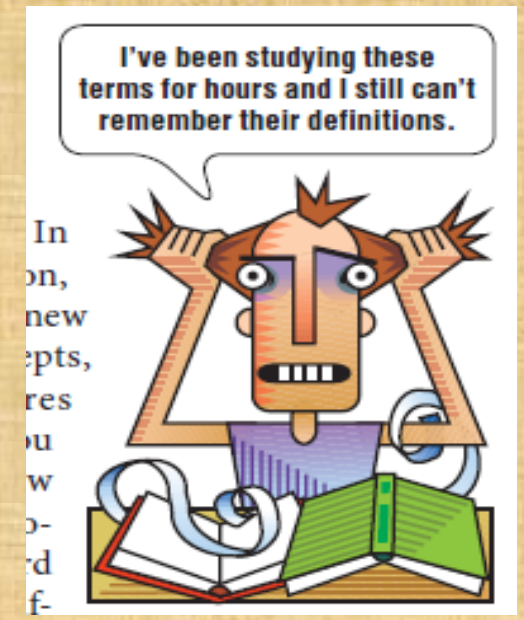
Cont …

- **Interest:** If a person has more interest, he will learn and retain better
- **Over learning:** Experiments have proved it will lead to better memory - repetition
- **Speed of learning:** Quicker learning leads to better retention – considering the pace of learners
- **Meaningfulness of the material:** Meaningful materials are better retained than for nonsense material
- **Sleep or rest:** Sleep or rest helps for clear memory and makes good connections in the brain

Meaning & Concepts of Forgetting



- Since our brain will ***be over exposed*** to various inf^{ns} daily, the tendency to ***remember decreases***
- Due to this we may forget the names of individuals, places …



Forgetting refers to the ***apparent loss*** of inf^{ns} already encoded and stored in the long-term memory

Theories of Forgetting

- **Decay theory**: memory traces fade with time if they are not accessed now and then.
  - i.e. due to ***simple passage of time***
- **Interference**: forgetting occurs due to particular memory ***interferes*** with the ***retrieval*** of other memories. ***Two types***:

= ***Proactive interference***: old memories interfere with new memories. **Old** → Interferes with → **New**

Cont …

= ***Retroactive interference***: material learned later disrupts retrieval of information learned earlier.

New → Interferes with → **Old**

- **Displacement theory/ New memory for old**
  - States that new information ***entering memory can wipe out old information***
  - Its mostly associated with STM – its capacity is 7± inf^{ns}
  - It ***cannot*** be associated with the LTM because of its ***virtually unlimited capacity***

Cont…

- **Motivation theory**: Freud claimed that we can forget experiences through **repression** –

  => Motivated forgetting i.e. through **repression**

  => People block those memories from ***consciousness*** that are too ***threatening*** or ***painful*** for them

  => Forgetting emotionally threatening experiences from consciousness such as:

  - witnessing a murder, Embarrassment, guilt and shock

- **Cue - dependency theory**: states that forgetting occurs due to **lack of adequate retrieval cues**.

Improving Memory

- **Pay Attention:**

  - Be conscious and pay attention

  - Then the encoded the information in the first place

- **Encode information in more than one way:**

  - The more elaborate the encoding, the more memorable

- **Add meaning:**

  - The more meaningful the material, the more likely it is to store in long-term memory

Cont …

- **Take your time:**

  - Minimize interference by using study breaks for rest or recreation.

  - Sleep is the ultimate way to reduce interference **- Rest**

- **Over learn:**

  - Rehearsal and repetition

- **Monitor your learning:**

  - Reviewing periodically, you will have a better idea of how you are doing